

Q18

Object A experiences a force F_A equal to Eq in the direction of the force. Starting from rest, its velocity at time t is $v_A = 0 + \frac{Eq}{m}t$. Hence its KE at time t is

$K_A = \frac{1}{2}mv_A^2 = \frac{1}{2}m\left(\frac{E^2q^2t^2}{m^2}\right) = \frac{E^2q^2t^2}{2m}$. In gaining this KE, it would have lost an equal amount of EPE.

Object B experiences a force F_B equal to $E(2q)$ in the direction opposite to that of the force. Starting from rest, its velocity at time t is $v_B = 0 + \frac{E(2q)}{2m}t = \frac{Eq}{m}t$. Hence its KE at time t is

$K_B = \frac{1}{2}(2m)v_B^2 = \frac{1}{2}(2m)\left(\frac{E^2 q^2 t^2}{m^2}\right) = \frac{E^2 q^2 t^2}{m}$. In gaining this KE, it would have lost an equal amount of EPE.

Comparing the two scenarios, object B would have lost twice the EPE compared to A at an instant after some time from rest.

Ans: D